

Nicolas Barrios

SOFTWARE ENGINEER · OPEN SOURCE SOFTWARE ENTHUSIAST

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Summary

Software Engineer and Open Source Contributor. With over 4 years of professional experience with modern C++ and Python implementing high performance software in the fields of high frequency trading and computer graphics, I'm interested in developing high-performance solutions and advancing emerging software platforms.

Work Experience

Simplex Trading

Chicago, IL USA

SOFTWARE ENGINEER

Oct 2023 - Feb 2026

- Contributed to a large C++ codebase, prioritizing system stability and long-term maintainability in a production-critical Linux environment.
- Navigated complex legacy systems to integrate new features and optimizations with minimal disruption to existing trading workflows.
- Built frontend applications using the Qt Framework to provide trading analysts with real-time visibility into the firm's trading systems
- Collaborated with trading analysts to improve existing systems, translating their quantitative requirements into production-ready software.

Bolt Graphics, Inc.

Remote, USA

SOFTWARE DESIGN ENGINEER (FULL-TIME)

Dec 2021 - Mar 2023

- Developed modular software variant of initial Bolt Graphics raytracing platform — C++
- Explored software-based raytracing best practices using publicly available documentation — Rust
- Implemented cross-platform build system integrating private IP libraries with existing rendering tools — CMake, vcpkg
- Applied CI/CD best practices on multiple cloud platforms, delivering software artifacts to company-wide teams — GitHub Actions, Azure DevOps

SOFTWARE SYSTEMS ARCHITECT (CONTRACTOR)

Aug 2021 - Dec 2021

- Explored a proof-of-concept implementation of Bolt Graphics IP, leading to provable results for IP success — NVIDIA OptiX

University at Buffalo School of Engineering and Applied Sciences

Buffalo, NY USA

TEACHING ASSISTANT

Aug 2019 - Dec 2021

- Assisted Professor Ethan Blanton in teaching the C-based Systems Programming course
- Hosted lab sessions where students implement short programs designed to show them the necessary systems programming skills
- Instructed additional after-hours sessions where students could lead course-adjacent Linux and C programming knowledge.
- Developed a introductory assignment to get students comfortable with C arrays and the C programming language as a whole

Projects

dirt - DNS Information Resolving Tool

GitHub Link

SOFTWARE ENGINEER

May 2023 - Ongoing

- An in-progress DNS resolver written in the Rust programming language
- My own implementation of the DNS standards RFC 1034 and RFC 1035
- Foray into networking by handling some of 'dig's simpler lookup functionalities.

Game of Life on GPGPU

GitHub Link

LEAD ENGINEER

Jan 2021 - May 2021

- Implemented Conway's Game of Life on NVIDIA's CUDA platform for GPGPU processing in C and C++
- Developed a parallel bitwise algorithms designed to concurrently compute a stencil over a large matrix.
- Measured a speedup of over 600x for the implemented algorithm, compared to the naive CPU implementation

Education

University at Buffalo, School of Engineering and Applied Sciences

Buffalo, NY USA

M.S. IN COMPUTER SCIENCE AND ENGINEERING - GPA: 3.57 / 4.00

Jan 2019 - Dec 2021

- Specializing in High Performance Computing and Parallel Processing

B.S. IN COMPUTER SCIENCE - GPA: 3.52 / 4.00

Aug 2017 - Dec 2021